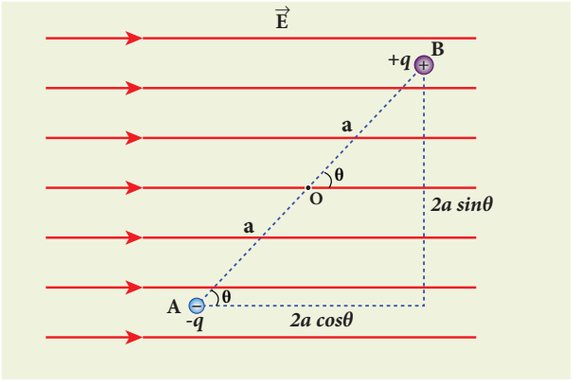



**Figure 1.31** The dipole in a uniform electric field

direction of the electric field. To rotate the dipole (at constant angular velocity) from its initial angle $\theta'$ to another angle $\theta$ against the torque exerted by the electric field, an equal and opposite external torque must be applied on the dipole.

The work done by the external torque to rotate the dipole from angle $\theta'$ to $\theta$ at constant angular velocity is

$$W = \int_{\theta'}^{\theta} \tau_{ext}\, d\theta \qquad (1.49)$$

Since $\vec{\tau}_{ext}$ is equal and opposite to $\vec{\tau}_E = \vec{p} \times \vec{E}$, we have

$$\left|\vec{\tau}_{ext}\right| = \left|\vec{\tau}_E\right| = \left|\vec{p} \times \vec{E}\right| \qquad (1.50)$$

Substituting equation (1.50) in equation (1.49), we get

$$W = \int_{\theta'}^{\theta} pE \sin\theta\, d\theta$$

$$W = pE\left(\cos\theta' - \cos\theta\right)$$

This work done is equal to the potential energy difference between the angular positions $\theta$ and $\theta'$.

$$U(\theta) - U(\theta') = \Delta U = -pE\cos\theta + pE\cos\theta'$$

If the initial angle is $\theta' = 90^\circ$ and is taken as reference point, then $U(\theta') = pE\cos 90^\circ = 0$.

The potential energy stored in the system of dipole kept in the uniform electric field is given by

$$U = -pE\cos\theta = -\vec{p}\cdot\vec{E} \qquad (1.51)$$

In addition to p and E, the potential energy also depends on the orientation $\theta$ of the electric dipole with respect to the external electric field.

The potential energy is maximum when the dipole is aligned anti-parallel ($\theta = \pi$) to the external electric field and minimum when the dipole is aligned parallel ($\theta = 0$) to the external electric field.

## EXAMPLE 1.16

A water molecule has an electric dipole moment of $6.3 \times 10^{-30}$ Cm. A sample contains $10^{22}$ water molecules, with all the dipole moments aligned parallel to the external electric field of magnitude $3 \times 10^5$ N $C^{-1}$. How much work is required to rotate all the water molecules from $\theta = 0^o$ to $90^o$?

### *Solution*

When the water molecules are aligned in the direction of the electric field, it has minimum potential energy. The work done to rotate the dipole from $\theta = 0^o$ to $90^o$ is equal to the potential energy difference between these two configurations.

$$W = \Delta U = U\left(90^\circ\right) - U\left(0^\circ\right)$$

From the equation (1.51), we write $U = -pE\cos\theta$, Next we calculate the work done to rotate one water molecule from $\theta = 0^o$ to $90^o$.

For one water molecule

$$W = -pE\cos 90^\circ + pE\cos 0^\circ = pE$$



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